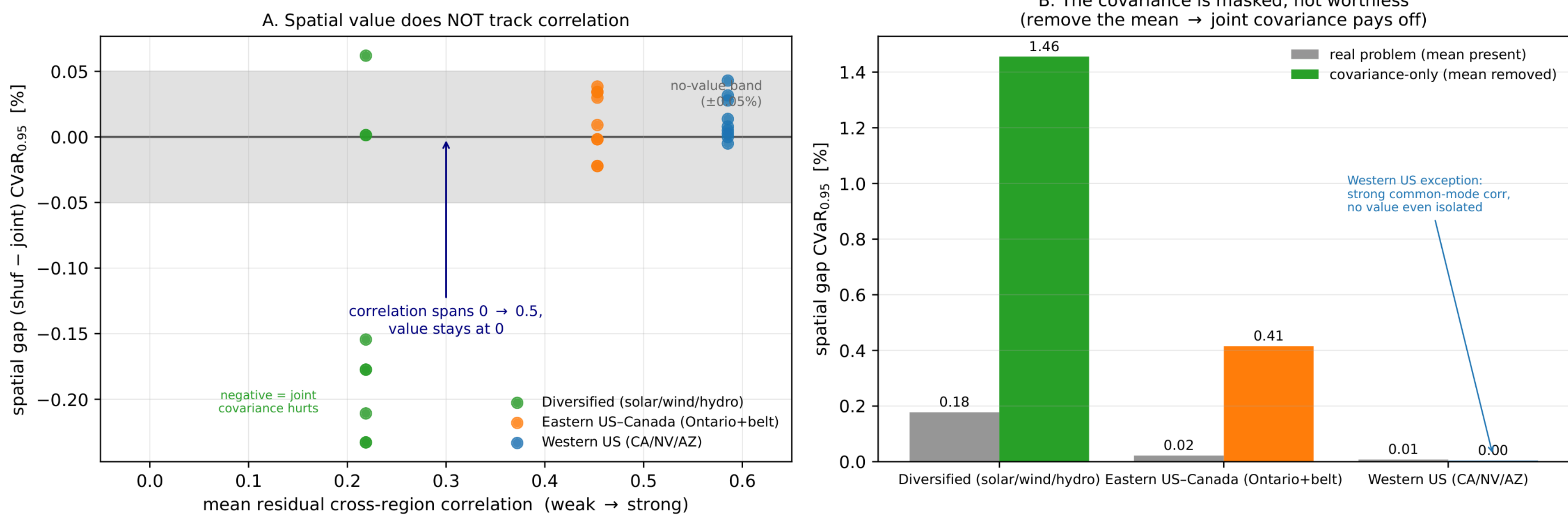


Does Spatial Correlation of Carbon Intensity Improve Robust Carbon-Aware Data-Center Scheduling?

A multi-grid falsification study: a causal mechanism, a copula test, and an active-transfer extension

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Phase 1: carbon correlation is real, but adds no robust scheduling value — because the mean field dominates



0.78

CROSS-GRID CORRELATION IS REAL

~0%

ROBUST SCHEDULING VALUE

+1.46%

ONLY WHEN THE MEAN IS FLATTENED

4–10%

ONLY VIA ACTIVE TRANSFER

The question

Compute is unusually **flexible**: batch and training jobs can shift in time and **across sites** (logically, over the network, no grid interconnection needed). Carbon intensity varies by hour and region, so a scheduler can defer work to cleaner moments under **distributionally robust optimization (DRO)**.

Prior models treat each region in isolation. We treat carbon intensity as a stochastic **vector** so that **spatial correlation** can inform the schedule, and ask whether it helps.

How we test it

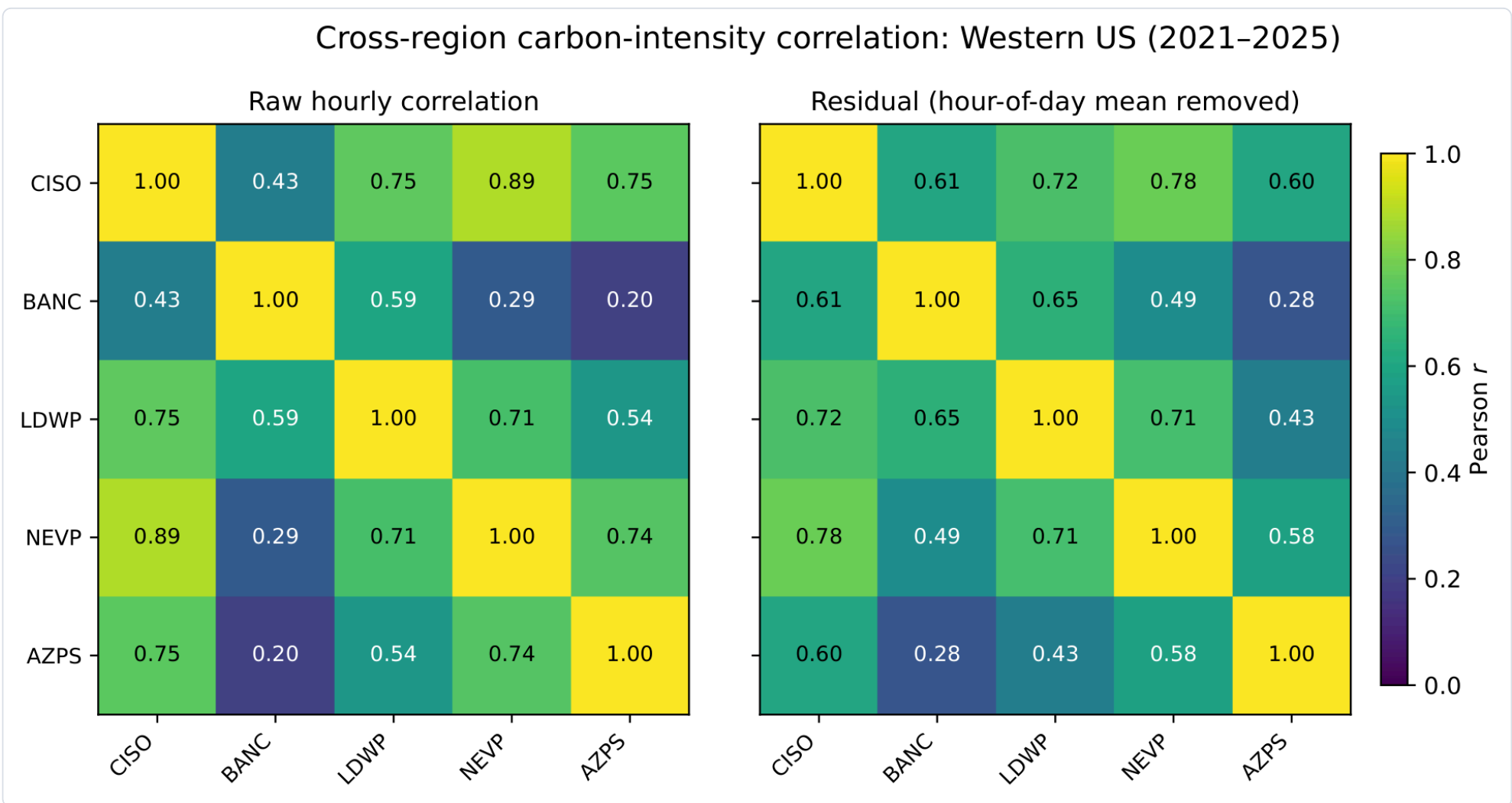
A Mahalanobis–Wasserstein DRO, solved as a second-order cone program:

$$\min_{x \in \mathcal{X}} \langle \hat{\mu}, x \rangle + \varepsilon \|L^\top x\|_2, \quad LL^\top = \hat{\Sigma}$$

Shuffled-marginals test. Fit the schedule on the *joint* covariance vs.\ a *block-diagonal* one with all cross-region structure zeroed, and compare out-of-sample CVaR_{0.95}. A positive gap means correlation helps.

- **Pre-registered:** config commit-locked, single 2025 test read.
- Three US/Canada grids spanning low-to-strong correlation ($\bar{r} \approx 0.2$ to 0.65).
- Best-case stress test: real fleets are dispersed (low correlation) and also null.

The premise holds



Seen directly: the joint and shuffled-covariance schedules coincide (US West)

THE FINDING

Spatial correlation of carbon is *real*, but a robust scheduler can’t use it.

Across every grid and every dependence model, the out-of-sample spatial gap never leaves **zero**. The value lives in the diurnal **mean** carbon field, not in how regions correlate.

It adds no robust value

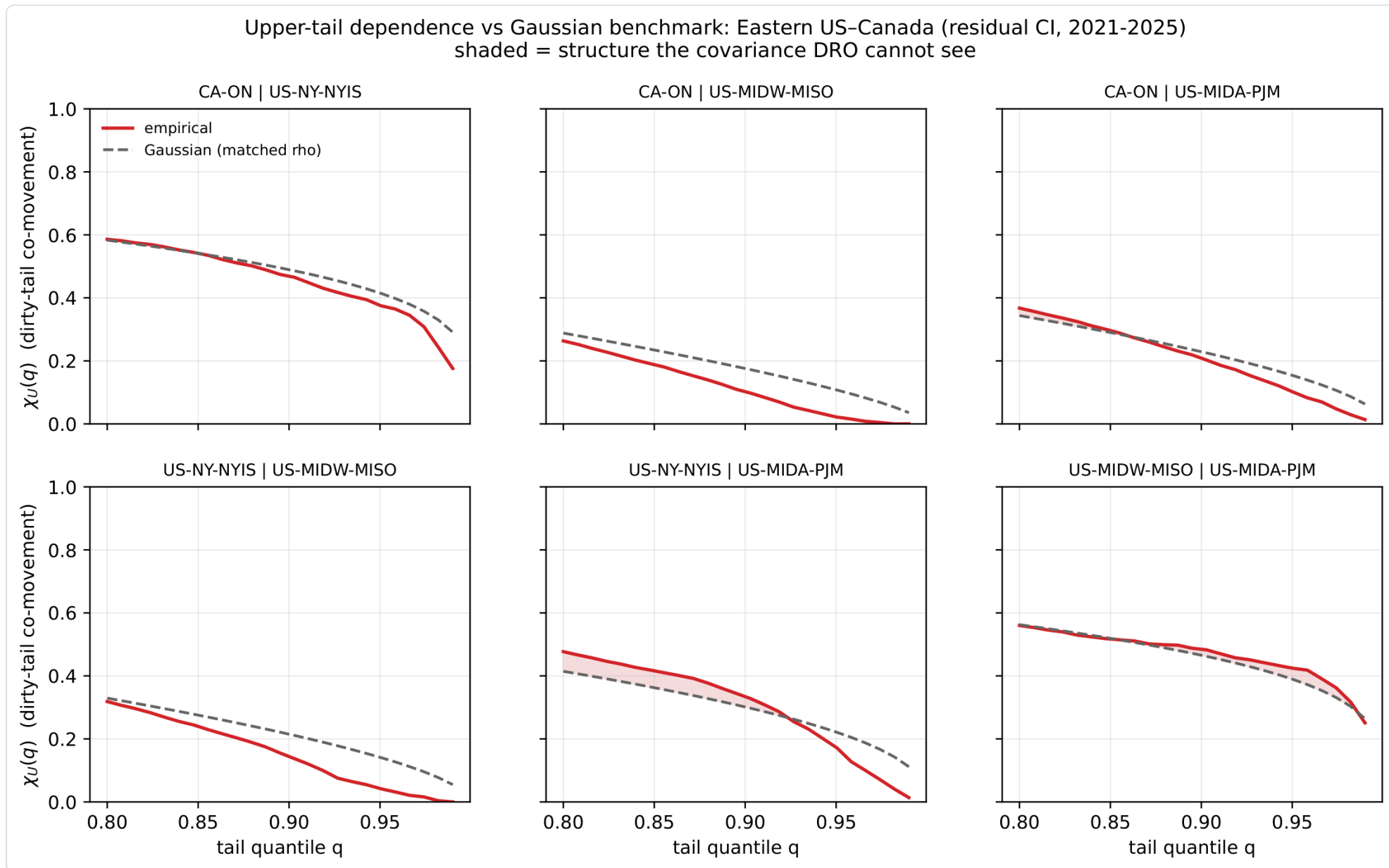
Across the spectrum the spatial gap **never exceeds a few tenths of a percent** and is not sign-stable. It survives shrinkage, residualization, Benjamini–Hochberg over 144 cells, walk-forward, tightness sweeps, and the full CVaR_{0.90}–CVaR_{0.99} tail range.

Statistically zero, not undetected. A per-cell equivalence test (TOST) rejects any effect above the materiality margin in all 27 cells. And the DRO is genuinely *active* ($\varepsilon^* = 1$ almost everywhere), it pays for robustness and the covariance still buys nothing.

Why, the mechanism

Mean-ablation (causal). Flatten the mean field and the joint covariance suddenly pays, up to **+1.46%**: the spatial signal is *real but masked* by the diurnal mean, not absent.

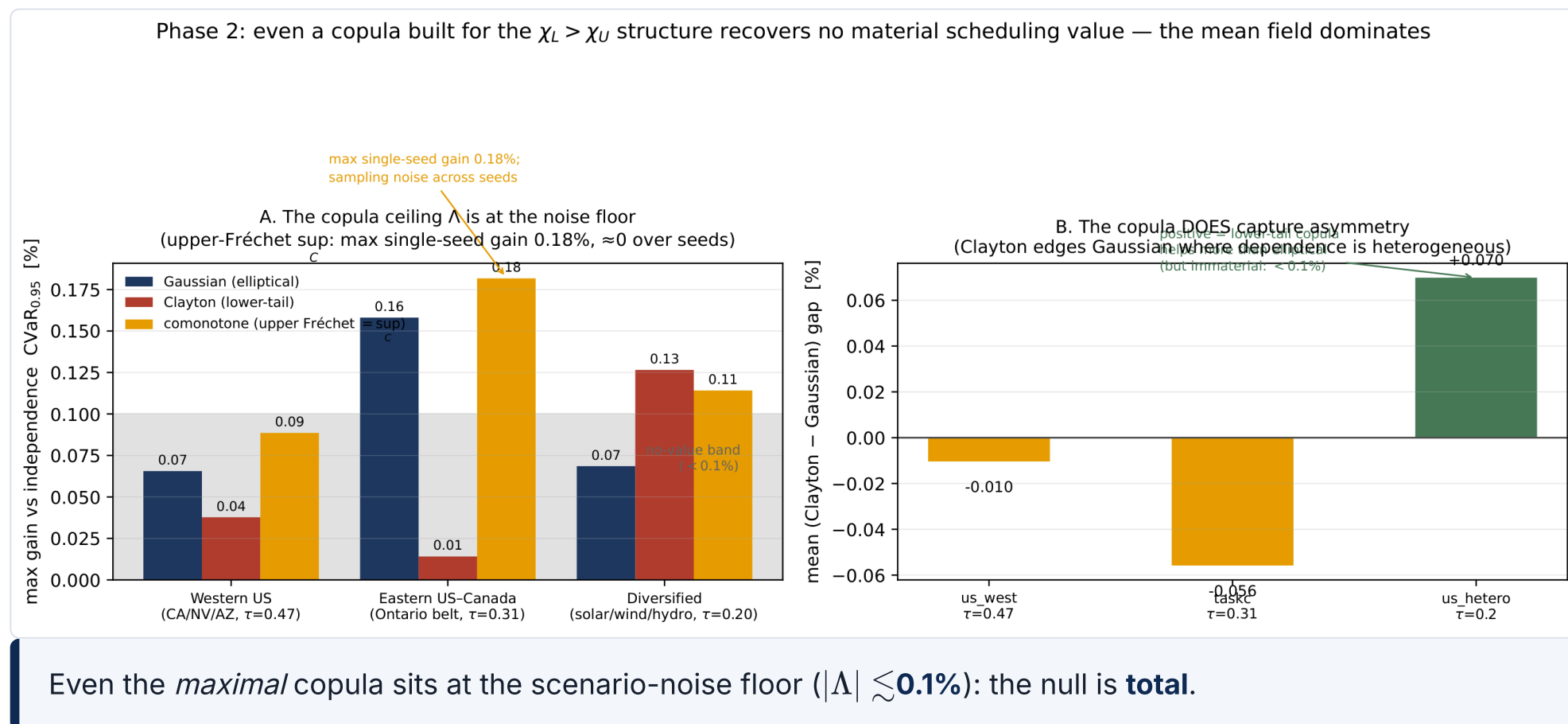
Tail dependence (structural). The residual dependence is *non-elliptical*: regions go clean together more than dirty together ($\chi_L > \chi_U$). An elliptical covariance ball forces $\chi_L = \chi_U$, it is the *wrong object*.



Clean-together exceeds dirty-together: the dependence a covariance ball cannot represent.

Even the right copula fails

We replace the covariance ball with copula schedulers that *can* see the asymmetry: independence, Gaussian, lower-tail **Clayton**, and the maximal **comonotone** ceiling.



A mean-dominance bound

By translation invariance of CVaR the dependence model moves only the residual term, and the spatial gap is bounded a-priori:

$$|\text{OPT}(\Sigma) - \text{OPT}(\Sigma^{\text{shuf}})| \leq \varepsilon \left(\max_x \|x\| \right) \|\Sigma^{\text{off}}\|^{1/2}$$

It vanishes with no robustification ($\varepsilon \rightarrow 0$) or no cross-structure. The tight ceiling is *measured* by the comonotone arm, an order of magnitude below the mean-driven value.

Conclusions & extension

- Spatial correlation is **real but adds no robust value**, for any passive dependence model.
- The binding constraint is the **mean field** (bound + comonotone ceiling).
- **Use a per-region marginal scheduler**: it captures essentially all the value.
- An **active transfer channel** recovers **4–10% deterministically**, but robustifying it pays nowhere data-grounded (online or under real emergencies).

Reproducible: 187 unit tests, pre-registered configs, equivalence tests, multi-seed stability, an independent review. AI used as a coding/drafting assistant under the author’s direction.

Joint vs.\ shuffled-covariance schedules are visually indistinguishable (Western US): the spatial null, seen directly. Load (bars) tracks the low-carbon hours.

